Trimester courses: tasks and milestones

12 November 2026 (updated for the gradebook and final-project routes added on 12 November). Everything needed for Trimester courses to work end to end: backend, frontend, and decisions. Owners: **BE** backend developer, **FE** web app frontend, **PM** product and operations.

1. The finding that shapes the plan

- **There are two backend lines, and the web app LMS screens were built against this repository line, not the team line.** This repository has its own LMS API. The team's line, the one Render serves, has a different and larger one: cohort-based paths, LiveKit video, Paystack payments, community, quizzes, assignments, certificates, institutions. I compared against the last copy of the team's branch I could read (the remotes are not reachable from here, so it may be older than what Render runs now).
- **Consequence:** the Trimester pages, scholarships, coding labs and live-class controls the app calls exist only in this repository's line. They return 404 on the live server. At the same time the live server has routes the app does not call yet (cohort curriculum, student dashboard, lesson progress, quizzes, assignments, LiveKit join).
- **Recommendation:** the team's line is the single source of truth (it matches the spec: LiveKit, Paystack, audit, institutions). Port the Trimester public pages and scholarships onto its Trimester and Cohort models, *keeping the response shapes the app already uses* so the public pages need no rewrite. Then move the learner screens to the team's LMS routes.

Area	This repository	Team's line (Render)
Public Trimester list, programme page, enrol	Yes	No (Trimester and Cohort models exist, no public routes)
Scholarships	Yes	No
Coding labs, practice workspaces, inquiries	Yes	No
Curriculum	GET /lms/curriculum (weeks and days)	GET /lms/cohorts/:cohortId/curriculum (modules, lessons, quiz, assignment)
Student dashboard	GET /lms/dashboard	GET /lms/dashboard/student
Live classes	YouTube stream, own chat, hand, people, start and end	LiveKit tokens, socket chat, polls, attendance, leave
Quizzes, assignments, grading	Simple	Attempts, answers, rubric, grading queue, gradebook with weights, final project, audit
Payments, certificates PDF, community	Not present	Paystack, certificates with QR, channels and posts
Module-by-module free courses, providers, categories	Gating and tasks here; no providers	Live on Render

2. Decisions needed first

#	Decision	Recommendation	Owner
D1	Which backend line is the source of truth, and which branch does Render deploy?	The team's line. Merge this repository's Trimester, scholarship, lab and practice work into it.	PM, BE
D2	Weekly path: /lms/curriculum or /lms/cohorts/:cohortId/curriculum?	The cohort path (it is in the spec). Add the 22-node fields to it.	PM, BE
D3	Is a Trimester paid, free, or both?	Decide now. Paid needs Paystack checkout (milestone 6).	PM
D4	Can tutors create courses of type trimester?	Admin only. Tutors keep regular and free.	PM
D5	Practice workspace path: /api/practice or /api/lms/workspaces?	Keep /api/practice for free practice. Add the cohort-linked path later for graded work.	PM, BE
D6	Does a class use only LiveKit, or YouTube Live as well?	LiveKit first. YouTube Live later as the backup.	PM

3. What counts as 'Trimester works'

- A visitor sees the Trimester programme, signs up, and enrols (or applies for a scholarship).
- The student sees the weekly path, opens lessons, watches video, takes quizzes and submits assignments.
- A tutor schedules and runs a live class; students join with video and audio, chat and answer polls; attendance records itself.
- A tutor grades work; the student sees grades and progress; at the end the student gets a certificate.
- An admin sets up the Trimester, cohort, weeks and tutors, and reviews scholarships.

Size key (one developer, rough): S up to 1 day, M 2 to 4 days, L 1 to 2 weeks. These are estimates to order the work, not commitments.

Milestone 0: Align and deploy

Nothing else can be tested on the live site until the two lines are one and the Trimester routes are deployed.

ID	Task	Owner	Size	Depends on
M0.1	Decide D1 and confirm which repo and branch Render deploys from.	PM, BE	S	-
M0.2	Port to the team's line, keeping response shapes: GET /lms/public/trimesters and /:id, POST /lms/trimesters/:id/enrol, scholarships (public list, apply, admin create, edit, publish, review), inquiries, code labs. Use the team's Trimester and Cohort models.	BE	L	M0.1
M0.3	Port the practice workspaces API (/api/practice) and the 1 MB body limit for code routes.	BE	S	M0.1
M0.4	Set Render environment: LiveKit URL, key and secret; Paystack keys; S3; mail; JWT. Run npm run catalog:import.	BE	S	M0.1
M0.5	Create the first real Trimester, cohort, weeks and tutor accounts through the admin routes. Publish.	PM, BE	S	M0.2
M0.6	Run the smoke scripts against Render and record the results (LMS, module gating, practice).	BE	S	M0.2, M0.3

ID	Task	Owner	Size	Depends on
M0.7	Update the API docs page with every new route and an example response.	BE	M	M0.2

Done when: GET /api/lms/public/trimesters returns the programme on Render, a test student can enrol, scholarships list, and the smoke scripts pass against the live URL.

Milestone 1: Student can enrol and open the course

The path from landing page to first lesson, on the team's routes.

ID	Task	Owner	Size	Depends on
M1.1	Public pages (/trimester, /scholarships) check against the deployed routes; fix any shape differences.	FE	S	M0.2
M1.2	Move the LMS service to the team's routes: /lms/me, /lms/dashboard/student, /lms/cohorts/:cohortId/curriculum, /lms/lessons/:id and the progress update. Send cohortId (the server picks the active one if omitted).	FE	L	M0.2
M1.3	Rebuild the student dashboard, weekly path and lesson screens on that data. Keep today's design.	FE	L	M1.2
M1.4	22-node trail: add nodeIndex, phase and per-node completion to the curriculum response (weeks 1 to 10 daily pattern, week 11 assessments, week 12 final nodes) and send an example.	BE	M	D2
M1.5	Build the trail screen from that example.	FE	M	M1.4
M1.6	Lesson rules on the server: 90 percent watched to complete, progress heartbeat.	BE	M	M0.2

Done when: A new student enrols, lands on the dashboard, opens week 1, plays a lesson, and the progress is still there after a refresh.

Milestone 2: Learning activities

Quizzes, assignments, comments and community on the team's routes.

ID	Task	Owner	Size	Depends on
M2.1	Quiz flow: overview, start attempt, save answers, submit, result.	FE	M	M1.2
M2.2	Assignments: list, open, submit; tutor grading queue and grade with rubric and feedback.	FE	M	M1.2
M2.3	Lesson comments with replies and likes.	FE	M	M1.2
M2.4	Community channels, posts, replies, reports.	FE	M	M1.2
M2.5	Notifications for new grades, replies and upcoming classes (events and a list).	BE, FE	M	M2.2

Done when: A student takes a quiz, submits an assignment, a tutor grades it, and the student sees the grade and feedback.

Milestone 3: Live classes (the largest gap)

The live server issues LiveKit tokens, but the app has no video client and calls routes the team's line does not have.

ID	Task	Owner	Size	Depends on
M3.1	Confirm the LiveKit server is running and the keys work. Test a real room with a tutor and a student.	BE	S	M0.4
M3.2	Add what the app needs and the team's line lacks: GET a single session, start and end (or confirm automatic status changes), participant list. Document the socket /live events for chat, polls and hand raise.	BE	M	M3.1
M3.3	Attendance fills itself from join, leave and LiveKit webhooks; enforce the 10 minute join window; keep the manual override.	BE	M	M3.1
M3.4	Recording: LiveKit Egress to S3 with states processing, available and failed. Replay route with access control. Replay does not count as attendance.	BE	L	M3.1
M3.5	Add livekit-client. Rebuild the class room: join with the token, video and audio, screen share for the host, leave, countdown and the join window.	FE	L	M3.1
M3.6	Chat over socket.io (already installed), history from GET .../chat.	FE	M	M3.2
M3.7	Polls: launch and close (tutor), vote with POST /lms/polls/:pollId/votes (student).	FE	S	M3.2
M3.8	Tutor tools: schedule, edit, cancel, attendance roster with override.	FE	M	M3.3
M3.9	Replay page for past classes.	FE	M	M3.4
M3.10	YouTube Live as the backup stream (spec Q6).	BE, FE	M	D6

Done when: A tutor and a student join a real class, see and hear each other, chat and vote in a poll, attendance is recorded, and the recording appears afterwards. (M3.10 can follow later.)

Milestone 4: Grading, final project and certificates

What turns attendance and work into a result. The backend side landed in the live docs on 12 November, so what remains is mostly the screens.

ID	Task	Owner	Size	Depends on
M4.1	**Done on the live server (12 Nov).** Gradebook with the seven weights (15, 15, 15, 10, 10, 10, 25), CSV export, transcript, override and configurable weights. **Still open:** late and resubmission caps are not in the docs.	BE	S	M2.2
M4.2	**Done on the live server (12 Nov).** Final project stages, stage submit with resubmission, tutor review and administrator approval. Confirm there are eight stages and send example responses.	BE	S	M4.1
M4.3	**Done on the live server (12 Nov).** Eligibility at 70 percent is returned by the gradebook and transcript; approval unlocks the programme certificate. Certificate PDF and public verification already exist.	BE	S	M4.1

ID	Task	Owner	Size	Depends on
M4.4	**Done on the live server (12 Nov).** GET grade-audits lists overrides and weight changes. Check it also records approvals and late-policy changes.	BE	S	M4.1
M4.5	Student progress and transcript screens on GET my-transcript.	FE	M	M4.1
M4.6	Final project screens for student, tutor and admin.	FE	L	M4.2
M4.7	Tutor gradebook view with weights, override and CSV export.	FE	M	M4.1

Done when: A student who meets the rules sees the final grade and receives a certificate that verifies publicly; every grade change is traceable.

Milestone 5: Admin and tutor tools

So the programme can be run without developers.

ID	Task	Owner	Size	Depends on
M5.1	Enforce tutor approval on the server: course creation and submit return 403 unless approvalStatus is approved.	BE	S	-
M5.2	Apply D4: only admins can create trimester-type courses.	BE, FE	S	D4
M5.3	Admin routes for Trimester, cohort, week setup, seats and status; confirm they are complete and documented.	BE	M	M0.2
M5.4	Admin screens: Trimester and cohort setup, scholarship review, enrolment, class scheduling, reports.	FE	L	M5.3
M5.5	Tutor screens: roster, announcements, grading queue.	FE	M	M2.2
M5.6	Scholarship document upload (applications) and reviewer notes.	BE, FE	M	M0.2

Done when: An admin creates a Trimester, adds a cohort and tutors, schedules classes and reviews a scholarship without touching the database.

Milestone 6: Payments and extras

Only if D3 says a Trimester can be paid for.

ID	Task	Owner	Size	Depends on
M6.1	Paid Trimester: create a Paystack checkout, webhook confirms and enrols, seat count updates, receipt.	BE	L	D3
M6.2	Checkout screens and the paid state on the Trimester page.	FE	M	M6.1
M6.3	Link coding labs into the Practice page; add and rename files in Practice.	FE	M	M0.3
M6.4	GitHub connect and commit-based submission (spec Q7).	BE, FE	L	-
M6.5	Server-side code execution: only after an independent security review, as the spec requires.	BE	L	Security review

Done when: A paying student is enrolled automatically after payment, and coding work can be linked to labs.

4. Order of work

- **Critical path to a first cohort:** M0, then M1, then M2 and M3 in parallel, then M5.1 and M5.4. That gives enrolment, learning, live classes and the admin tools. Recording (M3.4) and YouTube Live (M3.10) can follow while the first weeks run. The gradebook screens (M4.5 to M4.7) are needed before week 12 for the certificate.
- **Do not start FE work on M1.2 until M0.2 is deployed.** The shapes must be real, not guessed.
- **Backend and frontend can run in parallel** once D1 and D2 are settled: M3.2 with M3.5, M4.1 with M4.5.

5. Needed from the team

- **Decisions D1 to D6** in section 2.
- **LiveKit:** server URL, API key and secret, and confirmation that a test room works.
- **Paystack:** keys and the callback URL, if D3 is paid.
- **Real Trimester content:** programme name, weeks, lessons, quizzes, assignments, and the tutors who will teach.
- **Scholarship terms:** eligibility, dates and number of places.
- **An example response** for the cohort curriculum and the final project routes.
- **Access to the team's current branch** so both sides work from the same code. The copy I could read may be behind Render.